

EDA Insights Summary

The automated EDA report was generated using **Sweetviz** on the **Master_DF.csv** dataset. The dataset contains **42,239 records** with six numerical sensor features, one activity (Pattern) column, and one health (label) column. The report provides a clear overview of the dataset structure, feature distributions, correlations, and data quality.

The dataset is **moderately imbalanced**. The **Healthy** class contains **27,840 samples**, while the **Unhealthy** class contains **14,399 samples**. Although both classes have a good number of observations, the difference in class size should be considered while training machine learning models. Using a **stratified train-test split** will help maintain the class distribution during training and testing.

The feature distributions show that some sensor values are **approximately symmetric**, while others are **slightly right-skewed**. Histograms and density plots indicate that most observations are concentrated within a particular range, with a few larger values forming long tails. This suggests that the dataset contains natural variation in sensor readings collected during different physical activities.

The scatter plots, pair plots, box plots, and violin plots show that the **Healthy** and **Unhealthy** classes have some visible differences, but they also overlap in many regions. This means that a single feature may not be enough to separate the two classes. However, using all sensor features together should help machine learning algorithms learn the patterns more effectively.

The correlation analysis shows that some RSS features are **moderately to strongly correlated**. These features carry similar information, so there may be some redundancy in the dataset. Tree-based models can usually handle correlated features well, while linear models may benefit from feature selection or dimensionality reduction if required.

The profiling report also shows that the dataset has **no missing values** and **no significant duplicate records**, indicating good overall data quality. Box plots reveal the presence of some outliers, but these are likely caused by natural differences in human activities rather than errors in data collection. Therefore, the outliers should be examined carefully before deciding to remove them.

Overall, the dataset is clean and suitable for machine learning. Before building models, the numerical features should be **scaled or standardized** for algorithms such as Logistic Regression, Support Vector Machine(SVM) and K-Nearest Neighbors (KNN). The moderate class imbalance should be handled using **stratified sampling** and appropriate evaluation metrics such as Precision, Recall, F1-score, and ROC-AUC. These preprocessing steps will help improve the performance and reliability of the final classification model.